

Nondata-Aided I/Q Mismatch and DC Offset Compensation for Direct-Conversion Receivers

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Abstract—A digital-signal-processing technique is presented that can compensate for the frequency-independent inphase/quadrature (I/Q) mismatch and dc offset in zero-intermediate-frequency direct-conversion receivers. The proposed compensator consists of dc offset and I/Q mismatch estimators followed by a dc offset canceller and self-image suppressor. The estimators are nondata-aided (NDA) schemes that are derived by assuming that the desired signal is a random variable having an arbitrary circular symmetric distribution. The Cramér–Rao lower bounds (CRLBs) for NDA estimation are derived, and the characteristics of the estimators are analyzed in terms of mean-square errors. In particular, it is shown that for white Gaussian distributed signals, the accuracy of the estimates approaches the CRLBs as the number of observed samples increases. The characteristics of the proposed method as well as the validity of the analytical results are demonstrated through computer simulation.

Index Terms—Direct-conversion receiver, inphase/quadrature (I/Q) mismatch compensation, nondata-aided (NDA) estimation.

I. INTRODUCTION

THE zero intermediate-frequency (IF) direct-conversion architecture, employing an analog quadrature mixer to tune the bandwidth of interest to zero IF, is regarded as a suitable choice for a software-defined radio (SDR) receiver because of its flexibility and low power consumption [1]. This architecture, however, has several deficiencies resulting in performance degradation of digital radio receivers, among which the dc offset and inphase/quadrature (I/Q) mismatch are considered the most serious [2]–[4]. Unlike the dc offset which can be mitigated by the use of a suitable highpass filter, the I/Q mismatch imposes a stringent requirement on the receiver design since it generates a self-image interference over the desired signal band if not compensated. This is especially important for recent digital wireless systems that adopt high-level constellations because, in the presence of self-image interference, the signal-to-interference plus noise ratio (SINR) required for successful detection of transmitted symbols can hardly be attained. The impact of the I/Q mismatch is even more severe in multicarrier systems, such as orthogonal frequency-division multiplexing (OFDM), because

a subcarrier signal can be corrupted by its negative frequency subcarrier signal of a much higher strength.

Various digital-signal-processing (DSP) techniques have been developed to compensate for the I/Q mismatch, which includes local test-tone injection methods [5]–[7], data-aided (DA) methods using known training sequences [8]–[12], and nondata-aided (NDA) methods exploiting some statistical properties of the received signal [13]–[20]. Among the recent results, [12] presented a DA scheme for OFDM systems based on a channel smoothness criterion and showed that the I/Q mismatch compensation can be separated from the phase offset compensation; and [18] and [19] presented blind signal estimation schemes based on second-order circular-symmetry assumption. Despite their respectable performances with short training sequences, the usefulness of the DA methods is limited because of the inevitable increase in complexity of channel equalization and frequency synchronization at the receiver. On the other hand, the NDA methods are preferred to the DA methods because such an increase of receiver complexity is not involved. Among the recent NDA methods, the symmetric adaptive decorrelation (SAD) algorithm [13], which was originally proposed for noise cancellation with signal leakage based on the decorrelation principle and then was used to cancel the image-band signal in low-IF receivers [14], can be simply adapted to cancel the self-image in zero IF receivers if the desired signal is uncorrelated to its self-image [16].

In this paper, an NDA DSP technique is presented that can compensate for the frequency-independent I/Q mismatch and dc offset in zero-IF receivers focusing on circular-symmetric signals, and then the optimality of the estimators is investigated based on the closed-form expressions of the Cramér–Rao lower bounds (CRLBs) and analytical mean-square errors (MSEs) for the estimators. The dc offset and I/Q mismatch estimators are derived based on the assumption that the desired signal is a complex random process with a circular-symmetric distribution which is also assumed in [15]–[19], and then are revised to make them suitable for implementation. It is shown that when the desired signal is white Gaussian distributed, the MSEs of the proposed estimators approach the CRLBs as the number of received samples increases. In addition, computer simulation results show that the proposed technique can outperform the existing method.

The remainder of this paper is organized as follows. Section II presents the signal model. Section III derives the proposed technique, and Section IV presents some analytical results. In Section V, the accuracy of the analytical results as well as the performance of the proposed technique are demonstrated through computer simulations. Finally, Section VI is the conclusion.

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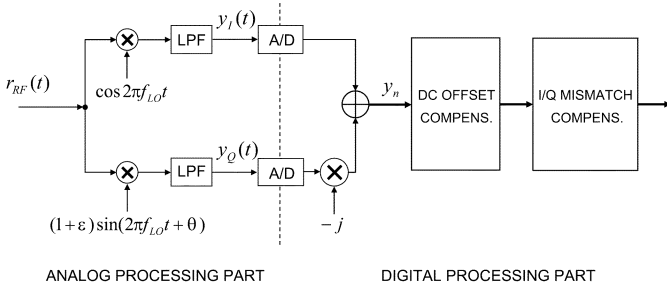


Fig. 1. System model.

II. SIGNAL MODEL

The system model considered in this paper is shown in Fig. 1. Here, $r_{RF}(t)$ is the received radio-frequency (RF) signal; f_{LO} is the local oscillator frequency; ϵ and θ denote the amplitude and phase mismatch of the quadrature mixer circuit; d_I and d_Q denote the dc offset components of I and Q branches, respectively.¹ The received RF signal can be represented as

$$r_{RF}(t) = 2\text{Re}\{s(t)e^{j2\pi f_C t}\} + w_{RF}(t) \quad (1)$$

where $s(t)$ is the baseband equivalent expression of the received signal, f_C is the carrier frequency and $w_{RF}(t)$ is the additive noise. In a direct-conversion receiver, a quadrature mixer is used to downconvert the RF signal to baseband, which then suffers from amplitude mismatch and phase mismatch as well as a dc offset. The mismatch parameters ϵ and θ are assumed to be quasistatic and frequency independent so that they are constant over an observation period and over a frequency band under consideration; and the dc offset d_I and d_Q are supposed to be quasistatic. Assuming an ideal filtering, the analog baseband signals before analog-to-digital conversion are given by

$$y_I(t) = \frac{1}{2} \left[x(t) + x^*(t) \right] + d_I$$

and

$$y_Q(t) = -\frac{1+\epsilon}{2j} \left[x(t)e^{-j\theta} - x^*(t)e^{j\theta} \right] + d_Q$$

where $x(t) = s(t)e^{j(2\pi\Delta f t + \phi)} + w(t)$; Δf is the carrier frequency offset given by $\Delta f = f_C - f_{LO}$; ϕ denotes the phase offset²; and $w(t)$ is the lowpass-filtered version of $w_{RF}(t)e^{-j2\pi f_{LO}t}$. Thus, the digital signal $\{y_n | n = 0, 1, \dots, N-1\}$, which is a sampled version of $y_I(t) - jy_Q(t)$ at the rate of $1/T$, can be expressed as

$$y_n = \beta_o x_n + \alpha_o x_n^* + d \quad (2)$$

where $\beta_o = (1/2)[1 + (1+\epsilon)e^{-j\theta}]$, $\alpha_o = (1/2)[1 - (1+\epsilon)e^{j\theta}]$, $d = d_I - jd_Q$, and $x_n = s_n e^{j(2\pi\nu n + \phi)} + w_n$; $\nu = \Delta f T$ is the normalized frequency offset, $s_n = s(t)|_{t=nT}$ and $w_n = w(t)|_{t=nT}$. Equation (2) illustrates the self-image interference phenomenon due to the I/Q mismatch in a direct-conversion receiver. The signal-to-noise ratio (SNR) of $\{x_n\}$ is defined by $E[|s_n|^2]/E[|w_n|^2]$.

¹If an analog dc offset canceller is employed before analog-to-digital converters (ADCs), then d_I and d_Q represent the residual dc offset.

²Phase offset represents the phase noise which occurs because of the local oscillator instability. In [12], the authors showed that their DA scheme can compensate the I/Q mismatch in the presence of phase noise.

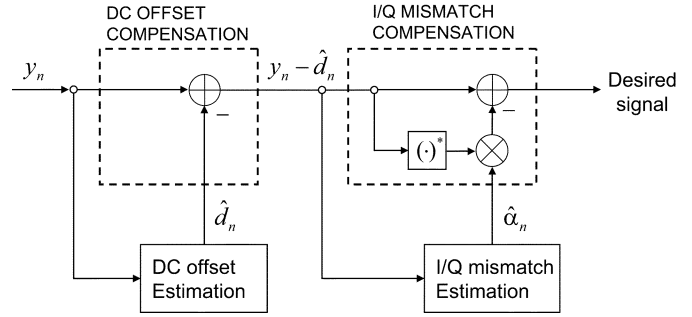


Fig. 2. Proposed compensation structure.

III. PROPOSED METHOD

In this section, the compensation structure and the proposed estimators of the dc offset and I/Q mismatch are derived.

A. DC Offset and I/Q Mismatch Compensation Structure

Given the received signal y_n in (2), the self-image interference and the dc offset can be suppressed by using the following relation:

$$(y_n - d) - \alpha(y_n - d)^* = (1 - |\alpha|^2) \beta_o x_n \quad (3)$$

where $\alpha = \alpha_o/\beta_o^*$. Notice that the right-hand side of (3) is a scaled version of the desired signal x_n ; thus, if α and d were known, the desired signal could be reconstructed by the left-hand side of (3) as shown in Fig. 2, eliminating the self-image of x_n and the dc offset. Accordingly, the I/Q mismatch and dc offset compensation problem reduces to the estimation of α and d .

B. Derivation of the Estimators

The unknown parameters α and d are derived as follows. Assume that $\{x_n\}$ is a zero-mean wide-sense stationary (WSS) random process whose distribution exhibits some circular symmetry (i.e., $E[x_n^2] = 0$), and ergodic in the wide sense.³ Then, from (2) and (3), we obtain

$$d = E[y_n] \quad (4)$$

and

$$E[(y_n - d) - \alpha(y_n - d)^*] = 0. \quad (5)$$

Solving (5) for α , we obtain

$$\alpha = \frac{b \pm \sqrt{b^2 - |c|^2}}{c^*}$$

where

$$b = E[|y_n - d|^2] = (|\beta_o|^2 + |\alpha_o|^2)E[|x_n|^2] \quad (6)$$

and

$$c = E[(y_n - d)^2] = 2\beta_o\alpha_o E[|x_n|^2]. \quad (7)$$

³A random process x_n is said to be ergodic in the wide sense [21], if $\lim_{N \rightarrow \infty} E[|(1/N) \sum_{n=0}^{N-1} f(x_n) - E[f(x_n)]|^2] = 0$, where $f(x_n)$ is given by x_n , x_n^2 , or $|x_n|^2$.

A feasible one between the two solutions of α is chosen as follows.

Lemma 1: $|\alpha| < 1$ for all practical values of $|\theta| < \pi/2$.

Proof: Using the definitions of β_o and α_o , we have $|\beta_o|^2 - |\alpha_o|^2 = (1 + \epsilon) \cos \theta$ and, thus, $|\alpha|^2 = 1 - (1 + \epsilon) \cos \theta / |\beta_o|^2$. This proves the lemma, because $\cos \theta > 0$ for $|\theta| < \pi/2$. ■

Lemma 2: Define $\alpha_1 = (b - \sqrt{b^2 - |c|^2})/c^*$ or $\alpha_2 = (b + \sqrt{b^2 - |c|^2})/c^*$. Then, $|\alpha_1| \leq 1$ and $|\alpha_2| \geq 1$.

Proof: α_1 and α_2 are conjugate-reciprocal pairs (i.e., $\alpha_1 = 1/\alpha_2^*$), meaning that they cannot exist inside or outside the unit circle of the complex plane at the same time. Now, let f and g be complex functions on (y_n, d) defined by $f = g^* = y_n - d$. Then, from the Cauchy-Schwartz inequality of $E[|f|^2]E[|g|^2] \geq |E[f \cdot g^*]|^2$, we have $b^2 - |c|^2 \geq 0$, which indicates that $|\alpha_1| \leq |\alpha_2|$ because b is positive real. Therefore, $|\alpha_1| \leq 1$ and $|\alpha_2| \geq 1$.

From the Lemmas 1 and 2, it is concluded that the feasible solution for α is given by⁴

$$\alpha = \frac{b - \sqrt{b^2 - |c|^2}}{c^*} \quad (8)$$

or alternatively

$$\alpha = \frac{c}{b + \sqrt{b^2 - |c|^2}}. \quad (9)$$

Now that d and α have been expressed as a function of the ensemble averages b and c , their estimates based on a single realization of the signal $\{y_n | n = 0, 1, \dots, N-1\}$ can be derived by exploiting the ergodic property of $\{x_n\}$ assumed before. Define

$$\hat{d}_N = \frac{1}{N} \sum_{n=0}^{N-1} y_n \quad (10)$$

$$\hat{b}_N = \frac{1}{N} \sum_{n=0}^{N-1} |y_n - \hat{d}_N|^2 \quad (11)$$

$$\hat{c}_N = \frac{1}{N} \sum_{n=0}^{N-1} (y_n - \hat{d}_N)^2. \quad (12)$$

Here, the subscript N represents the estimate that is based on N received samples. Then, as x_n is ergodic in the wide sense, so are y_n , y_n^2 , and $|y_n|^2$; thus, it follows that $\lim_{N \rightarrow \infty} \hat{d}_N = d$, $\lim_{N \rightarrow \infty} \hat{b}_N = b$, and $\lim_{N \rightarrow \infty} \hat{c}_N = c$.⁵ Consequently, if we define

$$\hat{\alpha}_N = \frac{\hat{c}_N}{\hat{b}_N + \sqrt{\hat{b}_N^2 - |\hat{c}_N|^2}} \quad (13)$$

we have $\lim_{N \rightarrow \infty} \hat{\alpha}_N = \alpha$ [22, p. 135].

The square-root operator in (13) can be realized by the use of a lookup table and a polynomial interpolator. However, the square-root operator in its current form has an implementation problem of requiring a large lookup table. It is because the dy-

⁴Both (8) and (9) are equivalent to the expression of α_1 in [17] and [20] when the image signal q_n is equal to the desired signal s_n (i.e., q_n is the self-image of s_n); An alternative derivation of (9) is also found in [19]. Notice that [19] as well as [17] and [20] focused only on the IQ mismatch neglecting the dc offset.

⁵The mean-square convergence of $\hat{b}_N$ and $\hat{c}_N$ to b and c follows because $\hat{b}_N = (1/N) \sum_{n=0}^{N-1} |y_n|^2 - |\hat{d}_N|^2$ and $\hat{c}_N = (1/N) \sum_{n=0}^{N-1} y_n^2 - \hat{d}_N^2$.

namic ranges of $\hat{b}_N$ and $\hat{c}_N$ depend on the given channel environment. To solve the problem, (13) is rewritten as

$$\hat{\alpha}_N = \frac{\hat{\gamma}_N}{1 - \mu_N} \quad (14)$$

where

$$\hat{\gamma}_N = \frac{\hat{c}_N}{2\hat{b}_N} \quad (15)$$

and

$$\mu_N = \frac{1 - \sqrt{1 - 4|\hat{\gamma}_N|^2}}{2}. \quad (16)$$

Note here that $|\hat{\gamma}_N|^2 \leq 1/4$ because $\hat{b}_N^2 \geq |\hat{c}_N|^2$, which can be simply shown by using the Cauchy-Schwartz inequality. Therefore, the term inside the square-root operator in (16) has its value between 0 and 1. ■

IV. ANALYTICAL RESULTS

In this section, the MSE bounds and the performance of the proposed estimator are analyzed. To make it mathematically tractable, it is assumed that $\{x_n\}$ is a zero-mean independent identically distributed (i.i.d.) complex Gaussian random process.

A. MSE Bounds for Unbiased NDA Estimators

Under the aforementioned assumption, $(y_n - d) - \alpha(y_n - d)^*$ is i.i.d. Gaussian distributed with the variance of

$$\sigma^2 = (1 - |\alpha|^2)^2 |\beta_o|^2 E[|x_n|^2]. \quad (17)$$

and, hence, the likelihood function of the unknown parameters $\vartheta = (\alpha, \alpha^*, d, d^*)^T$ for a given observation vector $\mathbf{y} = (y_0, y_1, \dots, y_{N-1})^T$ is written by

$$p(\mathbf{y}; \vartheta) = \frac{1}{(\pi\sigma^2)^N} \exp \left\{ -\frac{1}{\sigma^2} \|(\mathbf{y} - d\mathbf{1}) - \alpha(\mathbf{y} - d\mathbf{1})^*\|^2 \right\} \quad (18)$$

where $\mathbf{1}$ is the N -dimensional vector consisting of 1's. Recall that the CRLB, which assumes the regularity condition, is no greater than the Barankin bound which assumes no regularity condition [23]. Thus, the CRLB is meaningful as a lower bound whether the likelihood function satisfies the regularity condition or not. In this respect, assuming that $p(\mathbf{y}; \vartheta)$ satisfies the regularity condition for all $\vartheta_i \in \vartheta$, we derive the CRLB as a lower bound of the MSE. Let $\hat{\vartheta} = (\hat{\alpha}, \hat{\alpha}^*, \hat{d}, \hat{d}^*)^T$ denote an unbiased estimate of the vector ϑ (i.e., $E[\hat{\vartheta}] = \vartheta$). Then, it can be shown that [24]

$$E[(\hat{\vartheta} - \vartheta)(\hat{\vartheta} - \vartheta)^H] - \mathbf{M}^{-1} \quad (19)$$

is positive semidefinite, where $\mathbf{M} \in C^{4 \times 4}$ is the Fisher information matrix whose (i, j) th element is given by

$$\mathbf{M}_{i,j} = E \left[\left(\frac{\partial \ln p(\mathbf{y}; \vartheta)}{\partial \vartheta_i} \right)^* \frac{\partial \ln p(\mathbf{y}; \vartheta)}{\partial \vartheta_j} \right]. \quad (20)$$

Hence, from (19), we obtain

$$E[|\hat{\alpha} - \alpha|^2] \geq [\mathbf{M}^{-1}]_{1,1} \quad (21)$$

and

$$E[|\hat{d} - d|^2] \geq [\mathbf{M}^{-1}]_{3,3}. \quad (22)$$

After some calculations, it can be shown that

$$\mathbf{M} = \begin{pmatrix} N\mathbf{A} & \mathbf{0} \\ \mathbf{0} & \frac{N}{\sigma^2}(1 - |\alpha|^2)\mathbf{I} \end{pmatrix} \quad (23)$$

where $\mathbf{0}$ is the 2×2 null matrix, $\mathbf{I}$ is the 2×2 identity matrix, and $\mathbf{A}$ is the 2×2 matrix given by

$$\mathbf{A} = \begin{pmatrix} 2 + (N+1)|\alpha|^2 & (N+1)\alpha^{*2} \\ (N+1)\alpha^2 & 2 + (N+1)|\alpha|^2 \end{pmatrix}. \quad (24)$$

Therefore, using (23) and (24) in (21) and (22), we obtain

$$E[|\hat{\alpha} - \alpha|^2] \geq \frac{1}{2N} - \frac{(N+1)|\alpha|^2}{4N(1 + (N+1)|\alpha|^2)} \quad (25)$$

$$E[|\hat{d} - d|^2] \geq \frac{(1 - |\alpha|^2)|\beta_o|^2 E[|x_n|^2]}{N}. \quad (26)$$

The right-hand sides of (25) and (26) are the CRLBs of $\hat{\alpha}$ and $\hat{d}$, which will be denoted by CRLB_α and CRLB_d , respectively.

B. MSE Expressions for the Proposed Estimators

The bias and MSE of the dc offset estimator can be obtained as follows. Using (2) in (10), we have

$$\hat{d}_N - d = \frac{1}{N} \sum_{n=0}^{N-1} (\beta_o x_n + \alpha_o x_n^*). \quad (27)$$

Hence, we obtain $E[\hat{d}_N - d] = 0$ and

$$\begin{aligned} \text{MSE}_{\hat{d}_N} &= E[|\hat{d}_N - d|^2] \\ &= \frac{|\beta_o|^2 + |\alpha_o|^2}{N} E[|x_n|^2] + \delta_D \end{aligned} \quad (28)$$

where $\delta_D = (1/N^2) \sum_{n=0}^{N-1} \sum_{\substack{m=0 \\ m \neq n}}^{N-1} E[(\beta_o x_n + \alpha_o x_n^*)^* (\beta_o x_m + \alpha_o x_m^*)]$ is the autocorrelation term that is equal to zero for a zero-mean i.i.d. sequence $\{x_n\}$. In addition, it is a direct consequence of the central limit theorem [25], that if x_n is zero-mean i.i.d., then

$$\sqrt{N}(\hat{d}_N - d) = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} (\beta_o x_n + \alpha_o x_n^*) \quad (29)$$

converges in distribution to a random variable which is a zero-mean complex Gaussian with variance of $(|\beta_o|^2 + |\alpha_o|^2)E[|x_n|^2]$.

The asymptotic properties of the I/Q mismatch estimator are derived as follows. It is shown in Appendix A that if $\{x_n\}$ is zero-mean i.i.d., we have

$$\hat{b}_N = b + O_P\left(\frac{1}{\sqrt{N}}\right) \quad (30)$$

$$\hat{c}_N = c + O_P\left(\frac{1}{\sqrt{N}}\right) \quad (31)$$

which means that $\sqrt{N}(\hat{b}_N - b)$ and the real and imaginary parts of $\sqrt{N}(\hat{c}_N - c)$ are all bounded in probability. Using (30) and (31), $\hat{\alpha}_N$ can be expressed as (see Appendix B)

$$\hat{\alpha}_N = \alpha + \frac{|\beta_o|^2 - |\alpha_o|^2}{2\beta_o^{*2} E[|x_n|^2]} \left(\frac{1}{N} \sum_{n=0}^{N-1} x_n^2 + \hat{\mu}_x^2 \right) + o_P\left(\frac{1}{\sqrt{N}}\right) \quad (32)$$

where $\hat{\mu}_x = (1/N) \sum_{n=0}^{N-1} x_n$ and $o_P(1/\sqrt{N})$ represents the remainder term, indicating that $\sqrt{N}$ times the remainder converges to zero in probability as $N \rightarrow \infty$. If x_n is zero-mean i.i.d., $\hat{\mu}_x^2$ is $O_P(1/N)$ and is negligible, (32) can be rewritten as

$$\sqrt{N}(\hat{\alpha}_N - \alpha) = \frac{|\beta_o|^2 - |\alpha_o|^2}{2\beta_o^{*2} E[|x_n|^2]} \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x_n^2 + o_P(1). \quad (33)$$

Then, because it can be shown that $\{x_n^2\}$ is zero-mean i.i.d. if $\{x_n\}$ is i.i.d. and circular symmetric, it follows from the central limit theorem that the first term in the right-hand side of (33) converges in distribution to a zero-mean complex Gaussian random variable ξ with variance

$$E[|\xi|^2] = \frac{(1 - |\alpha|^2)^2}{4} \frac{E[|x_n|^4]}{E[|x_n|^2]^2}$$

which means that $\sqrt{N}(\hat{\alpha}_N - \alpha)$ in (33) also converges in distribution to ξ [21, p. 424]. Therefore, $\hat{\alpha}_N$ is asymptotically unbiased and its asymptotic variance is given by $1/N$ times the variance of ξ given by

$$\begin{aligned} \text{MSE}_{\hat{\alpha}_N} &= E[|\hat{\alpha}_N - \alpha|^2] \\ &= \frac{(1 - |\alpha|^2)^2}{4N} (\kappa_x + 2) \end{aligned} \quad (34)$$

where $\kappa_x = E[|x_n|^4]/E[|x_n|^2]^2 - 2$ is the kurtosis of the zero-mean circular symmetric complex random variable x_n .

Some remarks are of interest as follows.

- 1) From (28), the asymptotic MSE of the dc offset estimates normalized by the signal power (i.e., $E[|\hat{d}_N - d|^2]/E[|x_n|^2]$) monotonically decreases with N . It also varies with δ_D , which is a positive or negative real number depending on the correlation property of the given $\{x_n\}$.
- 2) Equation (34) indicates that the asymptotic MSE of the I/Q mismatch estimates also monotonically decreases with N , and increases with the Kurtosis of the signal $\{x_n\}$. In addition, the MSE of the I/Q mismatch estimates becomes smaller as $|\alpha|$ gets closer to 1, which means that the larger the I/Q mismatch becomes, the more accurate its estimates will be.
- 3) For a complex white Gaussian signal $\{x_n\}$, we have

$$\begin{aligned} \text{MSE}_{\hat{\alpha}_N} - \text{CRLB}_\alpha &= \frac{|\alpha|^2 - \frac{|\alpha|^4}{2} + \frac{(N+1)|\alpha|^2}{4(1+(N+1)|\alpha|^2)}}{N} \\ \text{MSE}_{\hat{d}_N} - \text{CRLB}_d &= \frac{2|\alpha_o|^2 E[|x_n|^2]}{N} \end{aligned}$$

which means that the MSE of the I/Q mismatch and dc offset estimator approach their corresponding CRLBs as $N \rightarrow \infty$ and $|\alpha| \rightarrow 0$.

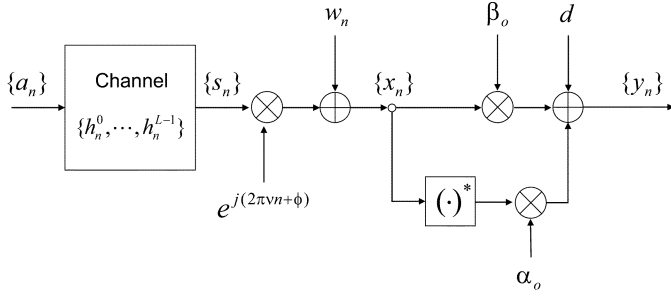


Fig. 3. Simulation model.

V. SIMULATION RESULTS

The performance of the proposed method was examined through computer simulation. Fig. 3 shows the signal model used in the simulation. The baseband equivalent signal in the desired band is expressed by $s_n = \sum_{l=0}^{L-1} h_n^l a_{n-l}$, where $\{a_n\}$ is the transmitted signal and $\{h_n^l | l = 0, 1, \dots, L-1\}$ denotes the zero-mean mutually independent L -tap channel responses at time n . Then, $x_n = (\sum_{l=0}^{L-1} h_n^l a_{n-l})e^{j(2\pi\nu n + \phi)} + w_n$, where $\{w_n\}$ is the AWGN.

The MSE for a complex Gaussian signal in AWGN channels was evaluated for the proposed estimator to confirm the analytical results in Section IV. The MSE performance was also evaluated for non-Gaussian signals in different channel environments to see whether the proposed estimator is viable in practical situations. In addition, the proposed algorithm was compared with the modified single-tap version of the SAD algorithm in terms of the image-rejection ratio (IRR). For fair comparison with the proposed method, a perfect dc offset canceller was assumed in the simulation of the SAD algorithm. The modified SAD algorithm for the compensation of the frequency-independent I/Q mismatch in zero IF receivers is summarized as follows: Let $w(0) = 0$. Then, the filtering and weight update equations are represented as $\hat{s}_n = y_n - w^*(n)y_n^*$, where the single-tap weight is updated by $w(n+1) = w(n) + \mu \hat{s}_n^{*2}$. Note that this method employs single-tap adaptive filters for the frequency-independent I/Q mismatch. It recursively estimates the I/Q mismatch parameter α by updating the tap weight $w(n)$.

The parameter values used in the simulation are as follows. The SNR defined in Section II was set to 20 dB. The normalized frequency offset $\nu = 0.01$ and the phase offset $\phi = 0$. The amplitude mismatch $\epsilon = 0.1$ and phase mismatch $\theta = 10^\circ$ ($\alpha_o = -0.0416 - j0.0955$, $\beta_o = 1.0416 - j0.0955$, and $\alpha = -0.048 - j0.0873$), and the dc offset was $0.1 + j0.1$. Four types of channels were considered: an AWGN channel, an eight-tap static Rayleigh channel, a flat fading channel, and an eight-tap frequency-selective fading channel having identical average power. The static Rayleigh channel coefficients were randomly generated at each trial to have a uniform power profile. For the Rayleigh fading channels, Jakes' model [26] was used for generating the channel parameters with $f_D T = 0.01$, where f_D is the maximum Doppler frequency. Two types of transmit signals were used: 1) complex white Gaussian signal and 2) an OFDM signal with 64 subcarriers and a cyclic prefix

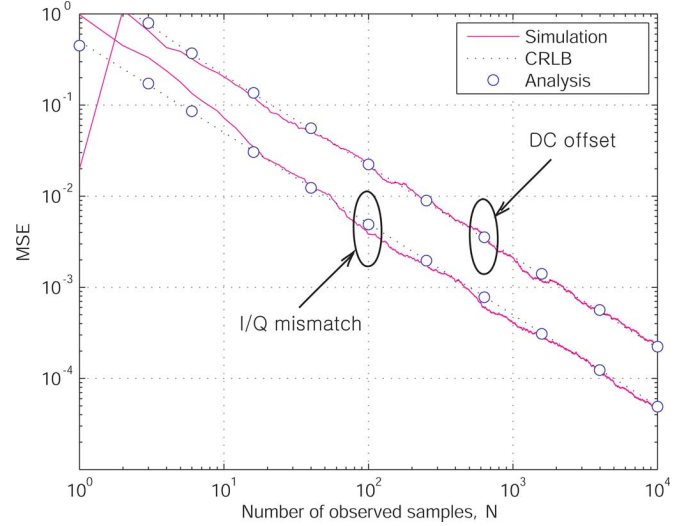


Fig. 4. MSE performances for complex Gaussian signals in AWGN channels.

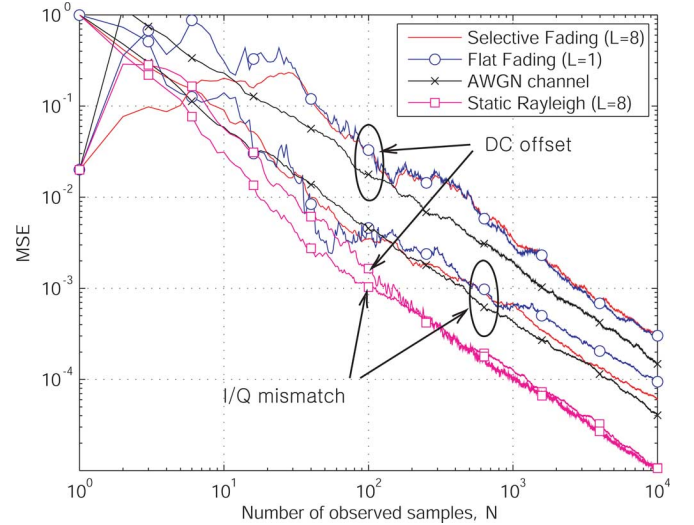


Fig. 5. MSE performances for OFDM signals.

length of 12. For simplicity, all subcarrier random data of the OFDM signal were QPSK-encoded. In the simulation, the MSE and IRR were empirically estimated based on 100 trials.

Fig. 4 shows the MSEs obtained from the analysis and simulation for the complex Gaussian signals in AWGN channels. A remarkably good agreement is observed for $N > 20$ between the analytical and simulation MSEs. As expected, the MSE decreases monotonically with N . In particular, the MSEs of the I/Q mismatch and dc offset estimates coincide with the CRLBs when $N > 20$.

Fig. 5 shows the simulation results obtained when the OFDM signals were used as the transmit signal $\{a_n\}$. Although the learning curves in fading channels exhibit some fluctuations with N , they tend to monotonically decrease with N . It is observed that the MSEs of the dc offset and I/Q mismatch estimates are insensitive to the frequency selectivity of the channel. In the figure, we find a large gap between the MSEs of dc offset estimates in fading channels and in the AWGN channel, but this

TABLE I
KURTOSIS (κ_x) AND δ_D OF THE OFDM SIGNAL

Channel Type	κ_x	δ_D
AWGN channel	-0.01	-4.3×10^{-4}
Static Rayleigh (L=8)	-0.076	-1.3×10^{-3}
Flat fading (L=1)	1.78	1.4×10^{-3}
Selective fading (L=8)	0.23	1.2×10^{-3}

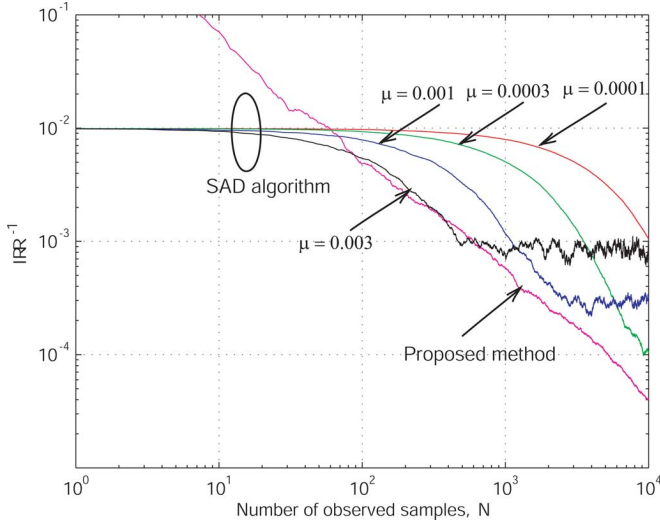


Fig. 6. Comparison of the IRR performances in AWGN channels.

gap had little impact on the MSE performance of the I/Q mismatch estimates because the dc offset estimation error was already sufficiently small. Equation (28) illustrates that this gap occurred because of the difference between the values of δ_D in the fading channels and AWGN channel, which was 1.2×10^{-3} as shown in Table I. As expected from (34), it is observed that for large N , the MSE of the I/Q mismatch estimates for different channels was determined by the order of magnitude of the Kurtosis of $\{x_n\}$ —in the flat fading channels where the Kurtosis of $\{x_n\}$ was greater than the others, the I/Q mismatch estimates exhibited the largest MSE. In addition, the proposed estimator showed the least MSE performance in the static Rayleigh channel. This occurs due to the small values of κ_x and δ_D as shown in Table I.

Fig. 6 compares the performance of the proposed method with that of the SAD algorithm modified for use in zero IF receivers. The IRR curve of the SAD algorithm was plotted for a set of step-size parameter μ . While the IRR curves of the SAD algorithm exhibit some floor as N increases, that of the proposed method monotonically decreases with N . In addition, it shows that the steady-state IRR of the SAD algorithm is inferior to that of the proposed method for all step-size parameter values.

The analog-to-digital converter resolution required by the proposed scheme was low; only a 5-bit ADC was sufficient to obtain the target MSE of 10^{-5} .

VI. CONCLUSION

In this paper, we presented a novel I/Q mismatch compensation method based on I/Q mismatch parameter estimation. Even though it was developed by assuming that the received signal is a wide-sense stationary random process having a circular symmetric distribution, the proposed method was shown to be robust even in nonstationary channels. It was also shown by analytical results and computer simulations that the learning curves of the proposed dc offset and I/Q mismatch estimators monotonically decrease with the number of observed samples, and the MSEs of both estimates in AWGN channels are remarkably close to the CRLBs. In addition, the proposed method has image-rejection performance superior to the conventional algorithm modified for use in zero IF receivers. Finally, it should be noted that the performance of the proposed scheme cannot be guaranteed for noncircular signals, such as binary phase-shift keying (BPSK) signals.

APPENDIX A DERIVATION OF (30) AND (31)

Using (2) in (11) and (12), we obtain

$$\hat{b}_N = \frac{1}{N} \sum_{n=0}^{N-1} |\beta_o x_n + \alpha_o x_n^*|^2 - |\hat{d}_N - d|^2 \quad (\text{A1})$$

$$\hat{c}_N = \frac{1}{N} \sum_{n=0}^{N-1} (\beta_o x_n + \alpha_o x_n^*) - (\hat{d}_N - d)^2 \quad (\text{A2})$$

where $\hat{d}_N - d$ is given by (27). Since the convergence in distribution of $\sqrt{N}(\hat{d}_N - d)$ noted below (29) implies $\hat{d}_N - d = O_P(1/\sqrt{N})$ [21] and $O_P(1/\sqrt{N})O_P(1/\sqrt{N}) = O_P(1/N)$, (A1) and (A2) can be rewritten by

$$\begin{aligned} \hat{b}_N &= \frac{1}{N} \sum_{n=0}^{N-1} (|\beta_o|^2 + |\alpha_o|^2) |x_n|^2 \\ &\quad + \frac{1}{N} \sum_{n=0}^{N-1} (\beta_o \alpha_o^* x_n^* + \beta_o^* \alpha_o x_n^2) + O_P\left(\frac{1}{N}\right) \end{aligned} \quad (\text{A3})$$

$$\hat{c}_N = \frac{1}{N} \sum_{n=0}^{N-1} (\beta_o^2 x_n^2 + \alpha_o^2 x_n^{*2} + 2\beta_o \alpha_o |x_n|^2) + O_P\left(\frac{1}{N}\right). \quad (\text{A4})$$

Since $\{|x_n|^2\}$ and $\{x_n^2\}$ are i.i.d., it follows from the CLT that $\sqrt{N}$ times the first terms in (A3) and (A4) converge in distribution to complex Gaussian random variables with their respective means of b in (6) and c in (7), meaning that $\hat{b} = b + O_P(1/\sqrt{N}) + O_P(1/N)$ and $\hat{c}_N = c + O_P(1/\sqrt{N}) + O_P(1/N)$. Since $O_P(1/\sqrt{N}) + O_P(1/N) = O_P(1/\sqrt{N})$, we obtain (30) and (31).

APPENDIX B DERIVATION OF (32)

Let c_I and c_Q ($\hat{c}_{N,I}$ and $\hat{c}_{N,Q}$) denote the real and imaginary parts of c ($\hat{c}_N$), respectively. Define two real-valued functions $f(\cdot)$ and $g(\cdot)$, so that $\hat{c}_N$ in (13) is decomposed as

$$\hat{c}_N = f(\hat{b}_N, \hat{c}_{N,I}, \hat{c}_{N,Q}) + jg(\hat{b}_N, \hat{c}_{N,I}, \hat{c}_{N,Q}).$$

Since their first derivatives exist at every point except on the line where $\hat{c}_{N,I} = \hat{c}_{N,Q} = 0$, $f(\cdot)$ and $g(\cdot)$ are analytic at a point (b, c_I, c_Q) and so they have a Taylor series at (b, c_I, c_Q) with the same radius of convergence of $|c|$.

Because $\hat{b}_N$ and $\hat{c}_N$ satisfy (30) and (31), $\hat{\alpha}_N$ around the point (b, c_I, c_Q) can be expanded as a Taylor series to the first order as [21, p. 423]

$$\hat{\alpha}_N = \alpha + (\hat{b}_N - b) \frac{\partial \hat{\alpha}_N}{\partial \hat{b}_N} + (\hat{c}_{N,I} - c_I) \frac{\partial \hat{\alpha}_N}{\partial \hat{c}_{N,I}} + (\hat{c}_{N,Q} - c_Q) \frac{\partial \hat{\alpha}_N}{\partial \hat{c}_{N,Q}} + o_P \left(\frac{1}{\sqrt{N}} \right) \quad (\text{B1})$$

where the derivatives at the point (b, c_I, c_Q) are evaluated by $\partial \hat{\alpha}_N / \partial \hat{b}_N = -\alpha / (|\beta_o|^2 - |\alpha_o|^2) E[|x_n|^2]$, $\partial \hat{\alpha}_N / \partial \hat{c}_{N,I} = 1 + \alpha^2 / 2 (|\beta_o|^2 - |\alpha_o|^2) E[|x_n|^2]$, and $\partial \hat{\alpha}_N / \partial \hat{c}_{N,Q} = j(1 - \alpha^2) / (2(|\beta_o|^2 - |\alpha_o|^2) E[|x_n|^2])$. Rearranging (B1)

$$\hat{\alpha}_N = \alpha + \frac{-2\alpha(\hat{b}_N - b) + (\hat{c}_N - c) + \alpha^2(\hat{c}_N - c)^*}{2(|\beta_o|^2 - |\alpha_o|^2) E[|x_n|^2]} + o_P \left(\frac{1}{\sqrt{N}} \right). \quad (\text{B2})$$

Using (6), (7), (A1), and (A2), it can be shown that

$$-2\alpha b + c + \alpha^2 c^* = 0 \quad (\text{B3})$$

and

$$\frac{-2\alpha \hat{b}_N + \hat{c}_N + \alpha^2 \hat{c}_N^*}{\frac{(|\beta_o|^2 - |\alpha_o|^2)^2}{\beta_o^{*2}}} = \frac{1}{N} \sum_{n=0}^{N-1} x_n^2 + \left(\frac{1}{N} \sum_{n=0}^{N-1} x_n \right)^2. \quad (\text{B4})$$

Substituting (B3) and (B4) in (B2), we obtain (32).

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